High-Precision Flood Mapping From Sentinel-1 Dual-Polarization SAR Data

Yanping Qin, Xiaobin Yin, *Senior Member, IEEE*, Yan Li, Qing Xu, *Member, IEEE*, Lei Zhang, Peng Mao, and Xingwei Jiang

***Abstract*— Synthetic aperture radar (SAR), with its ability to function under any weather conditions and at any time of day, along with multipolarization and frequent revisit capabilities, plays a crucial role in flood monitoring. However, SAR images face challenges, such as coherent speckle noise, feature mixing, terrain undulation, and adverse weather, making flood monitoring difficult. To address these challenges, this article proposes a high-precision flood mapping method from Sentinel-1 dual-polarization SAR data. We begin by generating false-color images through polarization combination and apply them to a multiscale segmentation approach, overcoming the limitations of single-polarization scattering and effectively reducing speckle noise. Digital elevation model (DEM) and reference water datasets are integrated into the segmentation process to mask terrain shadowing and permanent water. To reduce feature mixing effects, the optimal SAR image with minimal feature mixing is selected for flood mapping using the Gaussian mixture model. In the subsequent two-step classification process, fuzzy sets of texture features are incorporated to assist in categorizing uncertain regions, further reducing interference from feature mixing and enhancing flood recognition accuracy. Additionally, integrating pixel- and object-level analyses minimizes errors caused by improper segmentation. The proposed method is compared with several well-established algorithms, and the results demonstrate that our method outperforms the others in flood mapping accuracy. Analysis of years of flooding on the Leizhou Peninsula shows that Sentinel-1 SAR has the potential to effectively monitor the occurrence and development of floods.**

***Index Terms*— Flood, fuzzy, Gaussian mixture model, polarimetric-SAR (PolSAR), segmentation.**

I. Introduction

TYPHOONS, among the most devastating natural events, bring intense rainfall that often results in severe flooding in coastal regions. The occurrence of floods poses significant risks to the economy, agricultural production, and the safety and livelihoods of local communities [1], [2]. Consequently, real-time and efficient flood monitoring and early warning systems are of paramount importance. Due to its efficiency and extensive coverage, remote sensing has emerged as a key technique for flood mapping. The primary remote sensing imagery used for flood monitoring consists of active synthetic aperture radar (SAR) and passive optical images. Passive optical imagery provides rich spectral information for flood monitoring but is highly vulnerable to cloud interference [3], [4]. Thick clouds severely obstruct surface details, rendering optical imagery unreliable and significantly hindering its effectiveness in flood monitoring during heavy rainfall emergencies. Compared to optical images, SAR has advantages such as all-day, all-weather functionality, enabling effective flood monitoring under adverse weather conditions [5], [6]. Specifically, Sentinel-1 SAR serves as an excellent tool for flood monitoring and disaster management due to its wide coverage, short revisit interval, and free data access.

SAR systems demonstrate significant advantages in flood disaster monitoring, primarily attributable to the fundamental mechanism of distinct microwave backscattering characteristics between water and land surfaces. Typically, rough land surfaces generate high backscattering signals through diffuse reflection, while open water exhibits low backscattering due to specular reflection. However, the heterogeneous scattering mechanisms in complex environments substantially constrain the differentiation capability between flood and nonflood. In vegetated areas, the double-bounce reflection mechanism at vegetation–water interfaces significantly enhances backscattering intensity [7], whereas in urban regions, multiple reflection effects induced by architectural structures lead to anomalous scattering signal amplification in floodplains [8], [9]. To address scattering confusion in vegetated areas, multidimensional approaches have been proposed. Dasgupta et al. [10] developed a neuro-fuzzy classification algorithm that enhances flood mapping robustness in complex terrain scenarios. Cian et al. [11] achieved precise identification of open water and shallow vegetated water areas through fusion of multitemporal SAR-derived normalized flood indices and short vegetation indices. Grimaldi et al. [12] attained over 80% overall accuracy (OA) in flood-vegetation mixed area classification using probabilistic binning and fuzzy logic algorithms. For urban scattering interference mitigation, Zhang et al. [13] proposed a joint model combining backscattering coefficient increments and interferometric coherence

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Yanping Qin, Yan Li, Qing Xu, and Peng Mao are with the Faculty of Information Science and Engineering/Sanya Oceanographic Institution, Ocean University of China, Qingdao 266005, China (e-mail: qinyanping@stu.ouc.edu.cn; ly5298@ouc.edu.cn; xuqing@ouc.edu.cn; mp@stu.ouc.edu.cn).

Xiaobin Yin is with the Faculty of Information Science and Engineering/Sanya Oceanographic Institution, Ocean University of China, Qingdao 266005, China, and also with the Laoshan Laboratory, Qingdao 266042, China (e-mail: yinxiaobin@ouc.edu.cn).

Lei Zhang is with the 91001 Unit of PLA, Beijing 100091, China (e-mail: zlei_best@hotmail.com).

Xingwei Jiang is with the SANYA Oceanographic Laboratory, Qingdao 266200, China, and also with the National Satellite Ocean Application Service, Beijing 100082, China (e-mail: xwjiang@mail.nsoas.org.cn).

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attenuation, effectively suppressing building-induced multiple reflections. Furthermore, Zhang and Hu [14] adopted a multiscale adaptive fusion strategy to achieve high-precision dynamic monitoring of flood inundation extent in complex urban settings. Notably, fully polarimetric-SAR (PolSAR) enables the effective separation of vegetation volume scattering, building double-bounce reflections, and water surface specular scattering through polarimetric decomposition, thereby improving classification accuracy [15], [16]. However, PolSAR systems face limitations in large-scale flood emergency monitoring due to data acquisition challenges, processing complexity, and hardware costs. In contrast, dual-polarization SAR systems demonstrate superior potential in terrain scattering characterization while maintaining high temporal resolution and cost-effectiveness through synergistic VV–VH or HH–HV channel observations. Recent studies indicate that pseudo-color composite techniques [14], [17], [18] and multipolarization feature fusion methods [19], [20], [21], [22] based on dual-polarization data can significantly optimize flood detection accuracy. Despite advancements in scattering mechanism analysis and texture feature extraction, the critical challenge remains in effectively coordinating different polarization features to mitigate scattering mechanism interference caused by complex terrain and surface cover heterogeneity.

In SAR image analysis, existing methods can be categorized into pixel- and object-level approaches according to information processing units. Pixel-level methods, operating on individual pixels through scattering intensity or polarization feature analysis, offer advantages in algorithmic simplicity and engineering implementation. However, their neglect of spatial contextual information renders them highly susceptible to speckle noise, often resulting in salt-and-pepper noise artifacts in classification outputs [23]. Although deep learning techniques partially suppress noise interference through end-to-end feature learning [23], [24], their inherent pixel-level processing nature still suffers from boundary ambiguity and insufficient spatial information utilization. Object-based image analysis (OBIA) overcomes these limitations by generating spatially homogeneous object units, enabling comprehensive analysis of geometric features, texture statistics, and topological relationships [25], [26]. Cai et al. [27] experimentally demonstrated its superior classification accuracy over pixel-level methods in heterogeneous vegetated areas. Leveraging its powerful spatial semantic expression capability, this technology has been widely applied in flood inundation mapping [13], [28], [29]. In OBIA, feature extraction and classification accuracy are fundamentally constrained by segmentation quality. Multiscale segmentation, while effective in handling spatial heterogeneity of geographical features [30], [31], remains limited by terrain complexity and parameter sensitivity, inevitably causing oversegmentation/undersegmentation [26]. The predominant undersegmentation induces erroneous aggregation of heterogeneous objects, propagating boundary ambiguity and significantly degrading classification accuracy.

In flood monitoring applications, threshold-based segmentation methods have gained popularity due to computational efficiency. Commonly employed techniques include Otsu's global thresholding [32], [33], [34], Kittler–Illingworth minimum error thresholding [35], and optimized local adaptive threshold algorithms for SAR data [36], [37], [38]. However, these methods demonstrate limited adaptability to spatial backscattering heterogeneity, particularly exhibiting significant classification accuracy fluctuations in complex scattering environments. Consequently, probability model-based optimization strategies have emerged as research foci. Giustarini et al. [39] introduced a probabilistic flood mapping framework with equal prior probabilities (0.5) for flood and nonflood regions, effectively extracting spatial distribution characteristics of flood events. The selection of statistical distribution functions proves critical in probability model applications [40], [41]. Gaussian mixture models, through multimodal distribution modeling, successfully represent mixed spectral and scattering characteristics of complex land cover types, having been applied to flood probability inversion [29], [42], [43], [44]. Additionally, gamma distribution demonstrates superior performance in fitting backscattering intensity probability density distributions of open water due to its asymmetric and right-skewed characteristics [45], [46]. Nevertheless, significant probability distribution overlaps among different land cover types persist due to coupled effects of speckle noise, mixed pixels, and complex scattering mechanisms, remaining a critical technical bottleneck requiring breakthrough in current probabilistic model applications.

To address the challenges mentioned above, this study proposes a novel flood mapping method based on SAR images. The main features and innovations of the method are as follows.

1) Fully utilize the dual-polarization SAR data. During the multiscale segmentation stage of the image, the dual-polarization combined image is employed for segmentation to apply target features more thoroughly. Following segmentation, the dual-polarization additive combination SAR image with the smallest mixing region is selected to complete the classification process, improving the accuracy and reliability of the mapping.
2) Two-step classification. Initially, the threshold-based approach is used to categorize SAR images into three classes: flood, uncertain region, and nonflood. Based on this preliminary information, the fuzzy logic classification method is applied to the uncertain regions, integrating grayscale values, and texture features of the SAR imagery. This approach effectively reduces the impact of feature mixing, significantly enhancing the classification accuracy of the flood mapping.
3) Object-pixel combined classification strategy. During the flood mapping process, the standard deviation of pixels within objects is analyzed to dynamically determine whether to switch from object-level methods back to pixel-level methods. This strategy not only effectively suppresses the interference caused by SAR speckle noise but also addresses common undersegmentation issues in traditional image segmentation methods. Through these improvements, the proposed method achieves substantial advancements in the accuracy and reliability of flood mapping.

The structure of this article is as follows. Section II delineates the study area and datasets. Section III provides a

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detailed explanation of the proposed methodology. Section IV presents the results and discussion. The conclusion of this article is drawn at the end.

II. Study Area and Datasets

A. Study Area

The Leizhou Peninsula, with an area of 13 225 km^2, is situated in the southwest part of Guangdong Province, bordered by the Gulf of Tonkin to the west and the South China Sea to the east [see Fig. 1(a)]. It experiences a tropical monsoon climate and lies along one of the primary typhoon tracks in the northwest Pacific Ocean. Approximately 25% of all typhoons that make landfall in Guangdong Province strike the Leizhou Peninsula. On average, 1.5 typhoons make landfall annually, with some years seeing up to 4, and 56% of these being particularly strong typhoons. The eastern coastal region of the Leizhou Peninsula is densely populated and boasts extensive arable land. The frequent occurrence of typhoons during the annual season poses significant challenges for coastal areas. In particular, local residents and agriculture are severely impacted by flooding caused by heavy rainfall. Therefore, the eastern coastline of the Leizhou Peninsula has been selected as the focus of this study, specifically examining the floods caused by typhoons that impacted the region between 2016 and 2021.

B. Sentinel-1 SAR Data

The Sentinel-1 mission comprises two satellites, Sentinel-1A and Sentinel-1B, forming a dual-satellite constellation that enables rapid product delivery and short revisit cycles of six days [47], [48]. Both satellites are equipped with C-band SAR sensors and operate in four imaging modes, each offering different characteristics: stripmap (SM), interferometric wide swath (IW), extra-wide swath (EW), and wave (WV). The data products are categorized into Level-0, Level-1, and Level-2. Level-1 products are available in two formats: single-look complex (SLC), which retains phase information but exhibits higher scattering, and ground range detected (GRD), which reduces scattering at the cost of losing phase information. Sentinel-1 also supports dual-polarization [49], enhancing its capability for feature identification. The Sentinel-1 data for the study area were obtained using the IW mode with VV and VH polarization. In this study, GRD data from the IW mode are used, and after preprocessing, the spatial resolution is 10 $\times$ 10 m, which satisfies the requirements for flood monitoring. From 2016 to 2021, a total of 19 SAR images of flood events in the study area were acquired from Sentinel-1. Table I provides the details of the SAR data used.

C. Auxiliary Datasets

The Sentinel-2 program includes two multispectral satellites, Sentinel-2A and Sentinel-2B, each offering a wide imaging bandwidth. Sentinel-2 delivers frequent and timely imagery, with a ten-day revisit cycle for a single satellite and a five-day cycle when both satellites are operational. Sentinel-2, equipped with an optical payload, captures 13 spectral bands, offering high-resolution data: four bands at 10 m, six bands at 20 m, and three bands at 60 m. For this study, we utilized only the four 10-m resolution bands: Band 2 (blue), Band 3 (green), Band 4 (red), and Band 8 (near infrared). We acquired Sentinel-2 optical images before and after the flooding event and combined them with Sentinel-1 SAR images to create a validation dataset. These datasets are employed to differentiate flooded from nonflooded regions, supporting the accuracy of our flood mapping analysis.

The Shuttle Radar Topography Mission (SRTM) was a collaborative effort between the National Aeronautics and Space Administration (NASA) and the National Geospatial-Intelligence Agency (NGA) [50]. Conducted aboard the Space Shuttle Endeavor from February 11 to 22, 2000, the mission lasted 11 days and employed radar interferometry, on-board and off-board antenna systems, and single data acquisition techniques to produce near-global land elevation data products. SRTM data are provided at two resolution levels: SRTM-1, with a spatial resolution of 30 m, and SRTM-3, with a resolution of 90 m. This research employs 30-m SRTM-1 digital elevation model (DEM) data as auxiliary data to correct errors caused by mountain shadows, which are often mistaken for water in SAR images [51], [52].

European Space Agency (ESA) has developed global land cover products with a 10-m resolution, utilizing data from both Sentinel-1 and Sentinel-2. These products include the WorldCover maps for 2020 and 2021 (v100 and v200, respectively) [53], [54], each generated using different algorithms. The WorldCover 2020 v100 product and the WorldCover 2021 v200 product achieve overall accuracies of 74.4% and 76.7%, respectively. The WorldCover products categorize land into 11 distinct types. Fig. 1(c) shows the WorldCover map of the research area. This study utilizes WorldCover data to extract cultivated areas within the study region, aiming to analyze the variations in backscatter coefficients under different vegetation cover types and polarization modes.

III. Methodology

The flood mapping method proposed in this study consists of three main parts, and the detailed process is shown in Fig. 2. First, a false-color image is created by combining the preprocessed dual-polarization data, and image segmentation is then performed using this image as a guide. The expectation–maximization (EM) algorithm is applied to estimate the GMM parameters for SAR data under VV, VH, and VV + VH polarizations and to compare the overlap ranges of GMM across different polarizations in order to select the optimal SAR image for flood monitoring. Finally, floods are extracted from the optimal SAR images using a two-step procedure to obtain the geographical distribution of flood damage in the study area.

A. Preprocessing

Sentinel-1 SAR data preprocessing includes multilook processing, speckle noise filtering, geometric correction, radiometric calibration, and conversion of backscatter coefficients. After the multilook processing is completed, the SAR images

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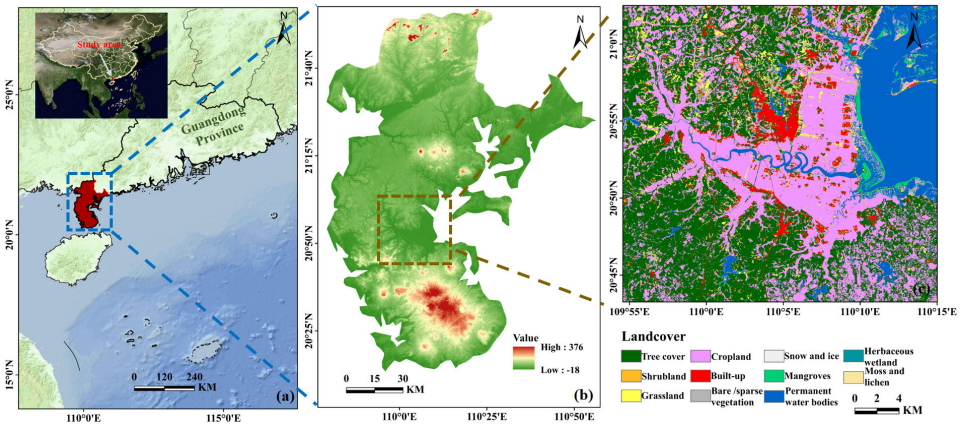


Fig. 1. (a) Geographic location of the study area, (b) Shuttle Radar Topography Mission 30-m digital elevation model data, and (c) WorldCover of the study area.

TABLE I
SAR IMAGES AFFECTED BY FLOOD

Number	Imaging time of SAR	Number	Imaging time of SAR	Number	Imaging time of SAR
1	2016/08/18	2	2017/07/20	3	2018/07/27
4	2019/07/28	5	2019/08/03	6	2019/08/09
7	2019/08/15	8	2019/08/21	9	2019/09/02
10	2020/07/28	11	2020/08/03	12	2020/08/09
13	2020/08/15	14	2020/08/21	15	2020/08/27
16	2021/07/17	17	2021/07/23	18	2021/08/04
19	2021/08/10				

are resampled to a regular grid with a resolution of 10 m. To reduce the impact of speckle noise while preserving edge details, the refined Lee filtering method is applied. For geometric correction, the distance-Doppler algorithm is used to reproject the images to the UTM Zone 36 S/WGS 84 coordinate system. Subsequently, the intensity values of the images are converted into backscatter coefficients through radiometric calibration. These dimensionless backscatter coefficients are then logarithmically transformed into decibel (dB) units. Data preprocessing is carried out with the Sentinel Application Platform (SNAP) software provided by the ESA. After completing the preprocessing steps, we normalize the backscatter coefficients to a grayscale range of 0–255 for further processing.

B. Comparison of Flood Mapping Across Land Covers and Polarizations

The flood regions within the study area are predominantly located in agricultural zones, where vegetation cover fluctuates over time. Changes in vegetation cover can significantly influence the backscattering coefficient of SAR, with the impact varying across different polarization modes. In this study, we examine how the backscattering coefficient of Sentinel-1 data differs across the two polarization modes under varying vegetation cover levels. Sentinel-1 and Sentinel-2 satellites can simultaneously observe the same area with a 7-h gap between observations, enabling the use of Sentinel-2 data to assess vegetation cover. Fig. 3 illustrates two examples of regions with dense vegetation cover, where the difference in backscattering coefficients between the vegetation-covered area and water is less pronounced in VV polarization compared to VH polarization. The difference arises because, with VV polarization, the electromagnetic wave interacts with the vertical structure of the crops, leading to a proportionate reduction in the scattered back electromagnetic wave. Consequently, relying solely on VV polarization data to extract water can result in a high commission rate. The higher accuracy Otsu thresholding approach is used in this article to validate the aforementioned findings. First, the threshold value is determined using the Otsu method and is then applied to compare the backscattering coefficients of each image under VV and VH polarization. The pixel is classified as water if the ratio of the backscattering coefficient to the Otsu threshold exceeds 1. Fig. 4 shows the ratio of the backscattering coefficient to the Otsu threshold for different polarization modes, corresponding to the SAR images in Fig. 3, along with the associated commission rates of water. The results indicate that the misclassification rate for VV polarization exceeds that of VH polarization, which is consistent with the previous conclusion.

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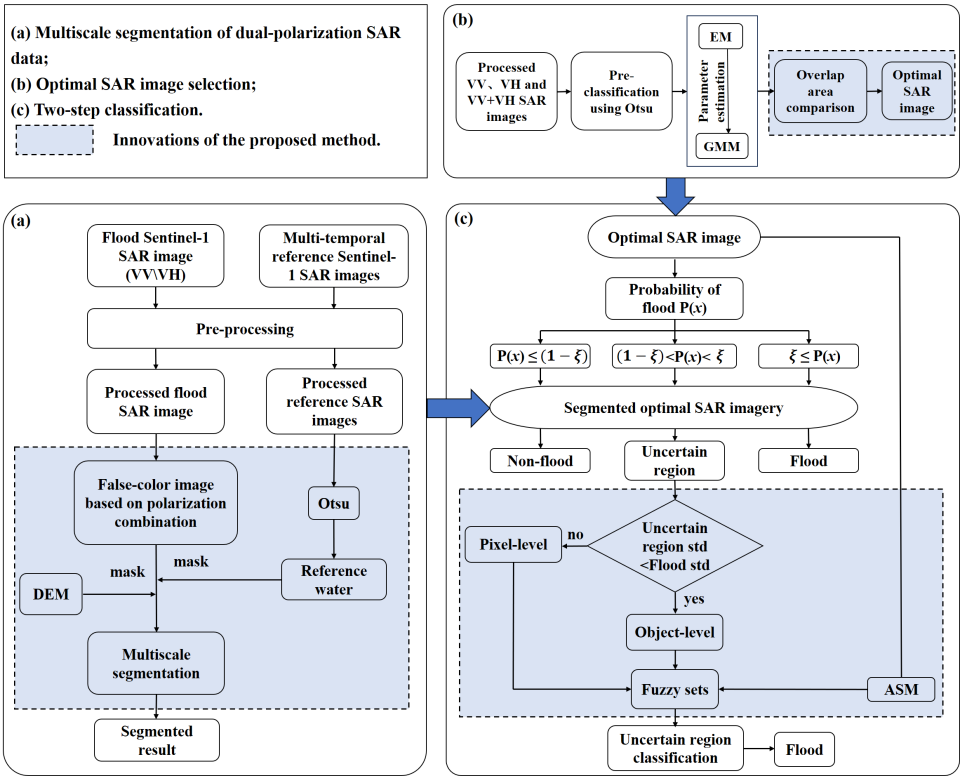


Fig. 2. Workflow for flood mapping. (a) Multiscale segmentation of dual-polarization SAR data, (b) optimal SAR image selection, and (c) two-step classification.

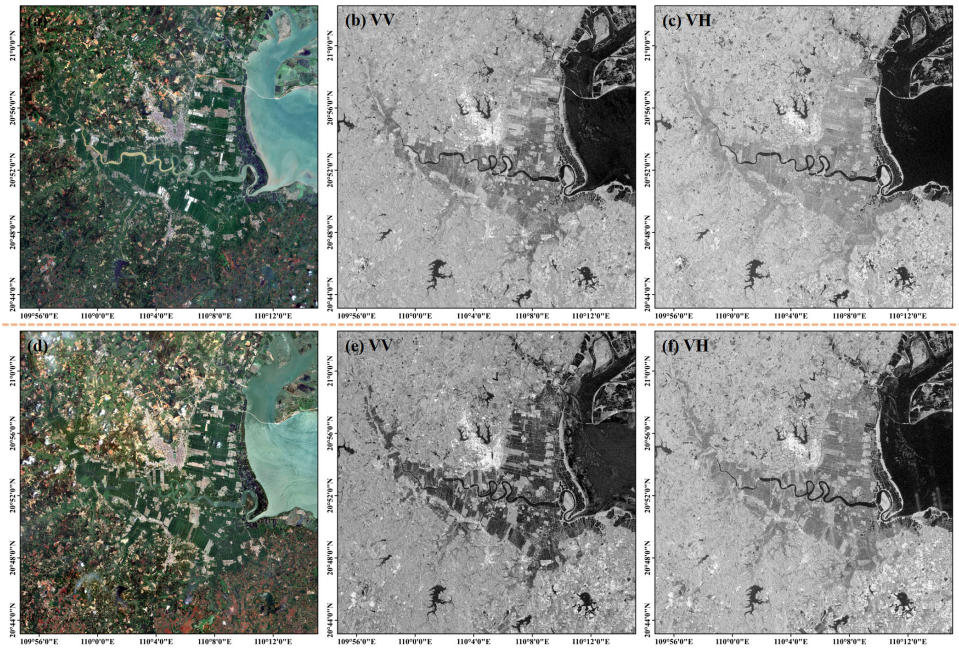


Fig. 3. Backscattering coefficients for various polarizations under dense vegetation. (a) and (d) Sentinel-2 multispectral imagery, (b) and (c) Co-temporal VV-polarized and VH-polarized SAR images corresponding to (a). (e) and (f) Co-temporal VV-polarized and VH-polarized SAR images corresponding to (d).

The backscattering coefficient is primarily affected by soil moisture and surface roughness when vegetation cover is minimal or absent. As shown in Fig. 5, two relevant examples illustrate that under sparse vegetation conditions, the difference in backscattering coefficients between water and cultivated land is smaller under VH polarization than under VV polarization. The difference between the two polarizations arises because the horizontal structure of the surface has a more significant impact on the backscattering coefficients of electromagnetic waves under VH polarization. The commission rate for water is determined by comparing the backscattering coefficients of each image with their respective Otsu thresholds. As shown in Fig. 6, under sparse vegetation cover, using only VH-polarized data results in higher false alarm

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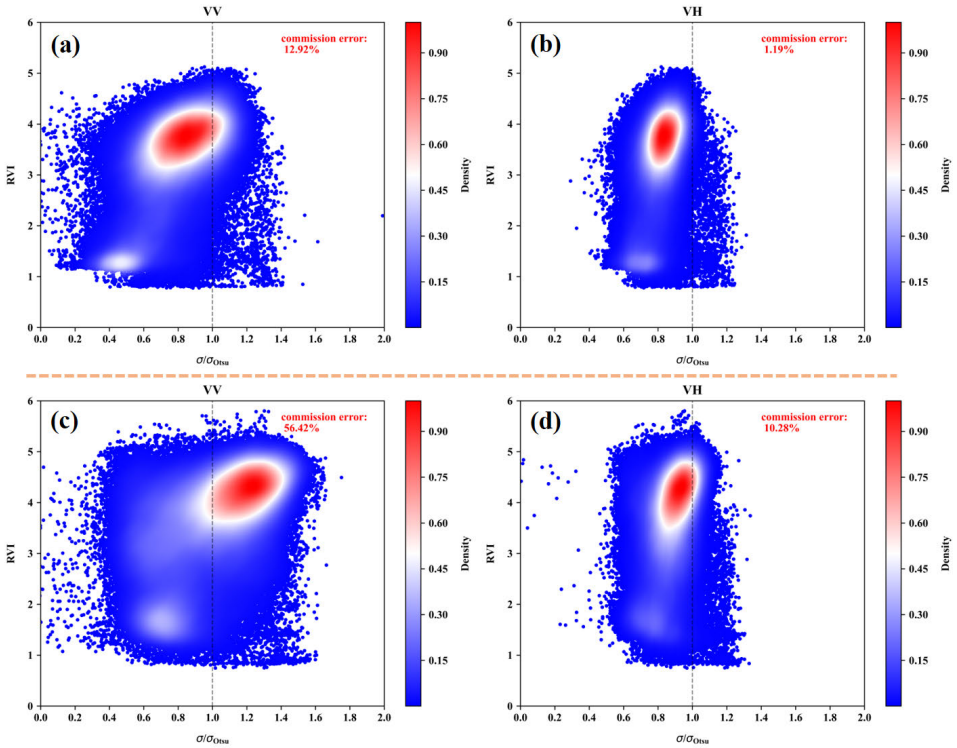


Fig. 4. Comparison of backscattering coefficients for each image pixel against the Otsu threshold at different polarizations under dense vegetation. (a) and (b) and (c) and (d) SAR images shown in Fig. 3(b) and (c) and (e) and (f), respectively.

rates. In conclusion, relying solely on one polarization mode for water extraction can lead to significant errors, as the backscattering coefficient is affected differently by various surface conditions and polarization modes. Therefore, a comprehensive approach that incorporates multiple polarization modes is essential for more accurate flood detection using SAR data.

C. Multiscale Segmentation

Image segmentation and object classification are the two main phases of object-oriented classification. Image segmentation, which divides an image into distinct regions based on its unique characteristics, serves as the foundation of classification, with its accuracy directly determining the precision of the classification results. In this study, ENVI 5.6 software is utilized for image segmentation, leveraging its ability to segment images based on pixel characteristics such as brightness, texture, and color. The software features a highly efficient edge-based segmentation algorithm that requires only a single input parameter to generate multiscale segmentation results, significantly improving both segmentation efficiency and accuracy.

As discussed in the previous section, in areas with complex surface cover, reliably identifying water from structures with similar backscattering characteristics is challenging when using single-polarization data alone, leading to a high commission rate. The combination of dual-polarization data overcomes the constraints of single-polarization data, providing more comprehensive scattering information and significantly improving the effectiveness of image segmentation. Therefore, this study integrates SAR dual-polarized data into the segmentation process by generating a false-color image that combines co-polarized and cross-polarized data. In the false-color image, the blue channel represents the co-polarized data, the green channel represents the cross-polarized data, and the red channel is assigned zero values. Due to the mountainous terrain in the study area, faint backscattering from distant mountains can be misclassified as water. Therefore, DEM data are incorporated during the segmentation process to mask and eliminate the influence of mountain shadows on water extraction results. Additionally, to accurately isolate flooded areas, reference water (areas that are permanently water-covered in the absence of flooding) must be precisely excluded. The reference water is derived using a thresholding method applied to multitemporal dual-polarization SAR data, and it is incorporated into the segmentation process as a masking operation. Fig. 7 presents two examples of multiscale segmentation under different polarization conditions. The scattering information under different polarization conditions is effectively fused in the false-color image, enabling the segmentation results to be more accurately based on the flood boundary.

D. Optimal SAR Image Selection

1) Gaussian Mixture Model: The GMM is a probability density function that assumes a sample is composed of several normal distributions with unknown parameters. The GMM can partition the entire dataset into multiple distinct distributions, making it a powerful tool for tasks such as land cover mapping. The land-flood segmentation problem is treated as a binary

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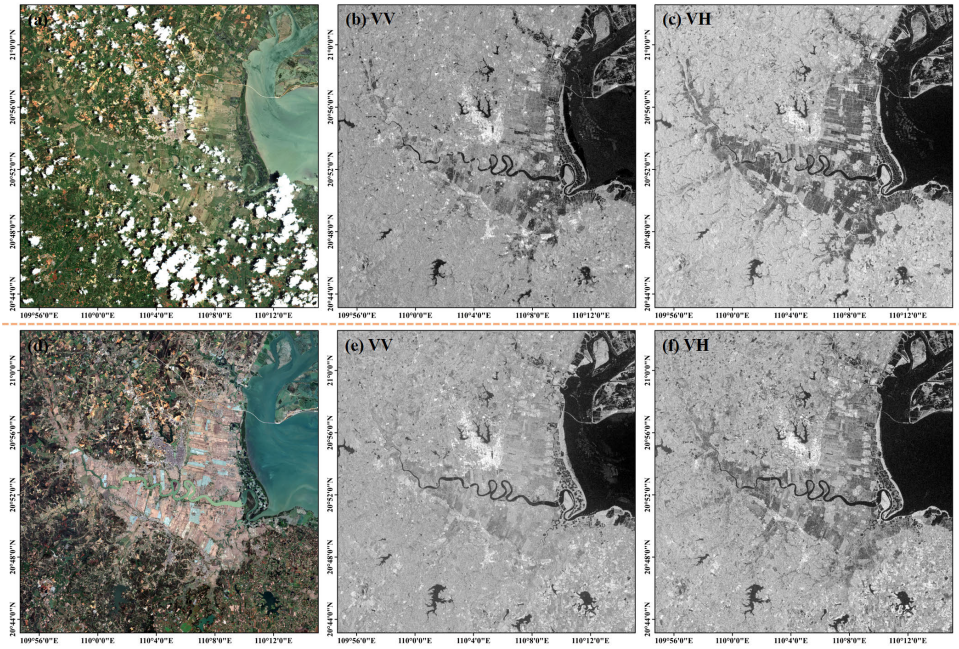


Fig. 5. Backscattering coefficients for various polarizations under sparse vegetation. (a) and (d) Sentinel-2 multispectral imagery. (b) and (c) Co-temporal VV-polarized and VH-polarized SAR images corresponding to (a). (e) and (f) Co-temporal VV-polarized and VH-polarized SAR images corresponding to (d).

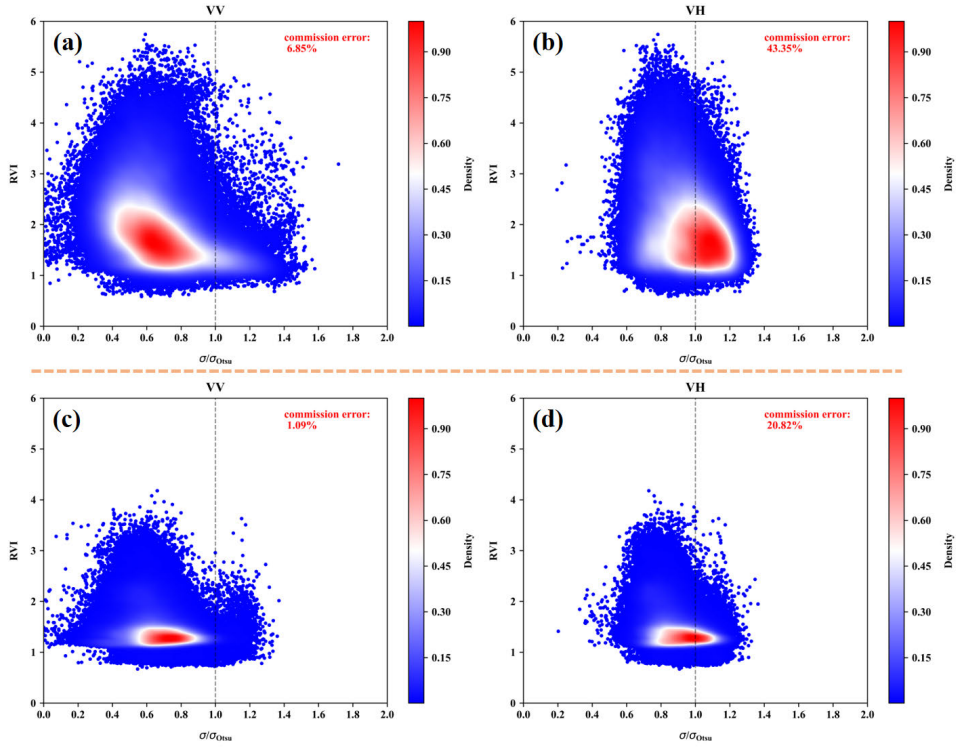


Fig. 6. Comparison of backscattering coefficients for each image pixel against the Otsu threshold at different polarizations under sparse vegetation. (a) and (b) and (c) and (d) SAR images shown in Fig. 5(b) and (c) and (e) and (f), respectively.

classification task, where the mixed distribution of flood and land pixels in SAR data is described by the following GMM:

$$p(x) = \alpha_1 N(x|\mu_1, \Sigma_1) + \alpha_2 N(x|\mu_2, \Sigma_2) \quad (1)$$

where x represents the gray value of an SAR image pixel; $p(x)$ denotes the probability of observing the gray value x; and $N(x|\mu_1, \Sigma_1)$ and $N(x|\mu_2, \Sigma_2)$ are the Gaussian probability density functions for the flood and nonflood distributions, respectively. The parameters a_1 and a_2 represent the probabilities that a pixel x belongs to the flood and nonflood distributions, respectively, with the condition that $a_1 + a_2 = 1$.

In (1), the parameters of the GMM are estimated using the EM algorithm. Initially, the Otsu thresholding method is applied to differentiate between flooded and nonflooded areas

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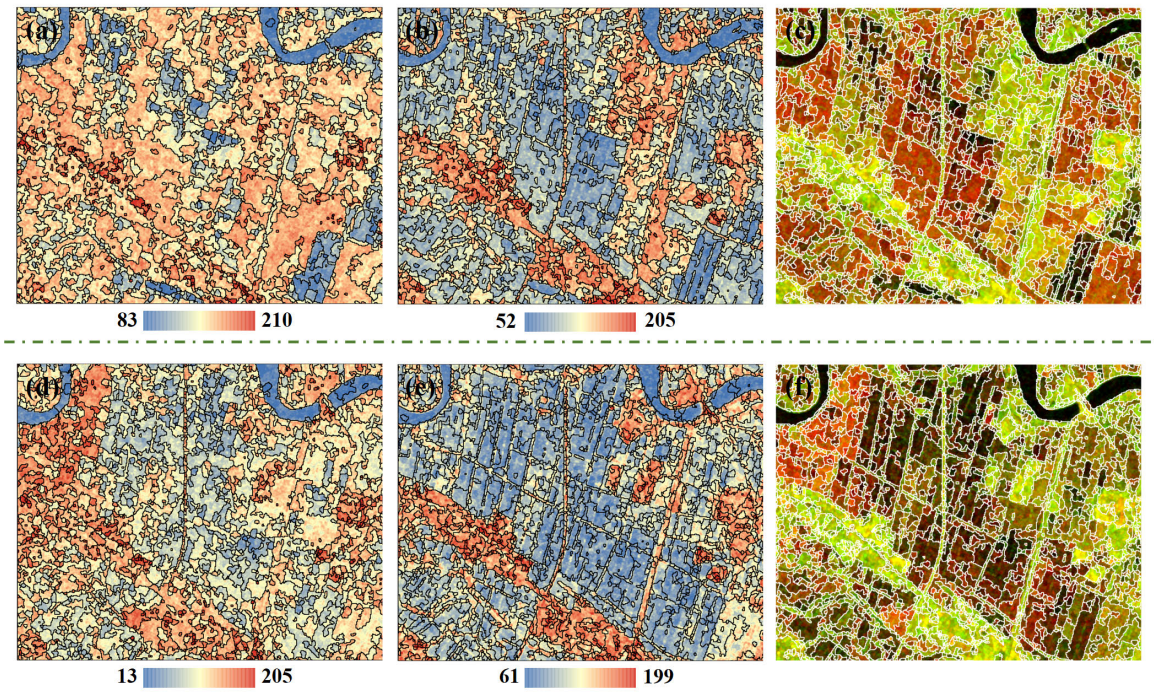


Fig. 7. Example of multiscale segmentation results under different polarization conditions. (a)–(c) VV, VH, and false-color segmentation results for SAR data from July 10, 2019. (d)–(f) Corresponding results from August 3, 2019.

in the image. Statistical data of these flood and nonflood areas are then computed based on the classification results and used as the initial parameters for the GMM. The GMM is subsequently iteratively optimized using the EM algorithm to determine the optimal model parameters.

2) Overlap Area Comparison: The probability distributions of flood and nonflood pixels fit by the GMM often overlap due to feature mixing and image noise. A larger overlap indicates that the distinction between flood and nonflood becomes more ambiguous in the feature space, making pixel classification more challenging and reducing the accuracy and reliability of the classification. To quantitatively evaluate the classification performance of SAR images under different polarization conditions and select the optimal image for classification, we calculate the overlap of the GMM for VV, VH, and VV + VH SAR images. A suitable threshold is essential for delineating the overlap region. To ensure that the gray values at the threshold primarily correspond to one distribution, the threshold should be sufficiently small, but not too small to avoid exaggerating the overlap. Our study shows that setting the distribution probability to 10^{-4} effectively strikes the balance. Accordingly, the higher gray value TH is determined when the probability p_{flood} ($p_{\text{flood}} = \alpha_1 N(x|\mu_1, \Sigma_1)$) in the flood distribution is 10^{-4}, while the lower gray value TL is calculated when the probability p_{nonflood}($p_{\text{nonflood}} = \alpha_2 N(x|\mu_2, \Sigma_2)$) in the nonflood distribution is 10^{-4}. Subsequently, the overlapping gray range is defined as $T = T_H - T_L$.

Fig. 8 displays the GMM fitting results for various polarizations, along with samples of the corresponding gray ranges for the overlap areas. Fig. 9 shows the mean overlap area under different polarization conditions for 2016–2021 SAR images. The results show that VV polarization, compared to VH, is more affected by environmental factors like wind and rain, resulting in higher backscattering coefficients over floods and larger GMM overlap areas, which reduce the accuracy of flood mapping. The observation is consistent with findings in the literature [29]. In certain cases, the VV-polarized SAR image can produce a histogram with only one visible peak [as shown in Fig. 8(c1)], which violates the assumption of a bimodal distribution. By combining both types of polarization data (VV + VH), the anti-interference capacity of the GMM is significantly improved, reducing the potential overlap areas. Consequently, the VV + VH combination consistently performs better in flood extraction tasks, making it the preferred choice for accurate and reliable flood mapping.

E. Two-Step Classification Approach

The optimal SAR images for flood mapping—VV + VH SAR images—are acquired in Section III-C. Even in the GMM of these optimal SAR images, overlap still exists between the flood and nonflood distributions. Relying solely on a threshold for classification can lead to underdetection or overdetection in certain regions, prompting the use of a two-step classification strategy in this study.

1) Initial Classification: Equation (2) is used to calculate the probability that each gray value corresponds to flood. The SAR images are then classified into three categories based on an appropriate probability threshold: flood, uncertain regions, and nonflood

$$P(x) = \frac{\alpha_1^{\text{opt}} N\left(x\middle|\mu_1^{\text{opt}}, \Sigma_1^{\text{opt}}\right)}{\alpha_1^{\text{opt}} N\left(x\middle|\mu_1^{\text{opt}}, \Sigma_1^{\text{opt}}\right) + \alpha_2^{\text{opt}} N\left(x\middle|\mu_2^{\text{opt}}, \Sigma_2^{\text{opt}}\right)} \tag{2}$$

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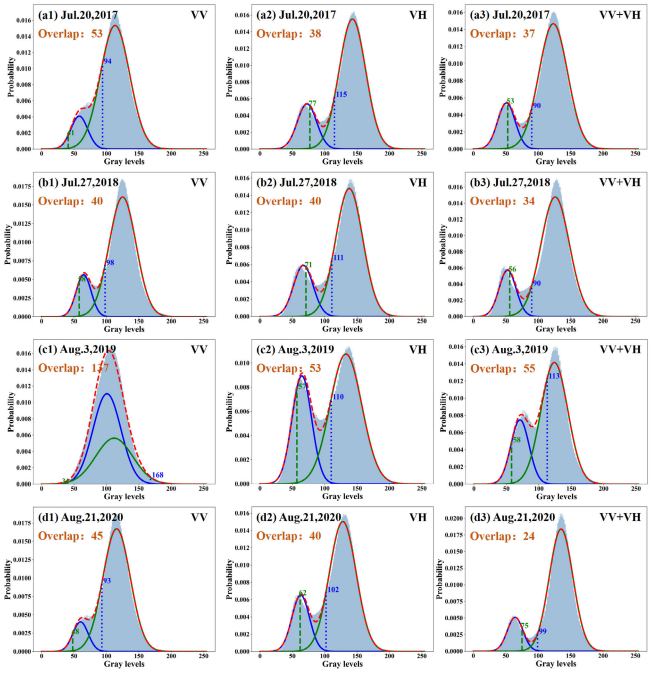


Fig. 8. Example showing the fit GMMs for VV, VH, and VV + VH polarizations, along with the corresponding overlap areas. The blue and green solid lines represent the probability density function of the fit components for the flood and nonflood regions, respectively. The red dashed line indicates the GMM fitting curve. The blue and green dashed lines denote the x-axis lines at the corresponding gray values when the probability densities of the fit components for the flood and nonflood regions are 10^{-4}. The corresponding gray values are also provided in their respective colors.

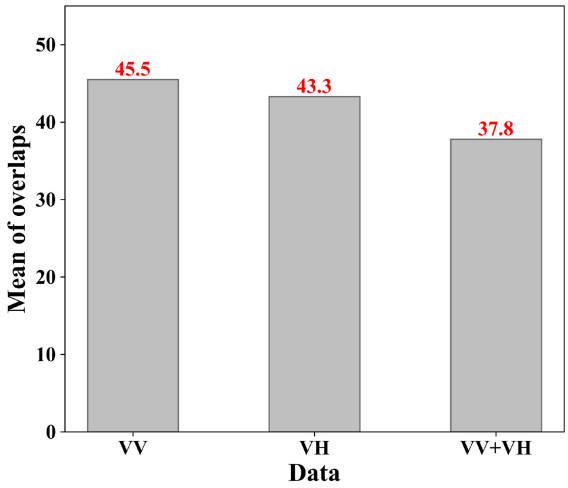


Fig. 9. Mean overlap area of the GMM under different polarization conditions for SAR flood images from 2016 to 2021.

where P indicates the probability of being the flood when the gray value is x, and $\alpha_1^{\text{opt}}, \mu_1^{\text{opt}}, \Sigma_1^{\text{opt}}, \alpha_2^{\text{opt}}, \mu_2^{\text{opt}}$, and Σ_2^{opt} are the optimal parameter of the GMM

$$\text{object category} = \begin{cases} \text{flood}, & \text{if } P(x) \geq \xi \\ \text{uncertain}, & \text{if } 1-\xi < P(x) < \xi \\ \text{nonflood}, & \text{if } P(x) \leq 1-\xi. \end{cases} \tag{3}$$

To determine the appropriate probability threshold, we gradually increase the threshold from 50% to 95% in 5% increments and assess the commission of flood and nonflood at each threshold. Here, we selected one SAR image from each year as representative data to evaluate the impact of the threshold on the classification results of flooded and nonflooded areas (see Table II). The analysis reveals that the commission of flood and nonflood is consistently below 1% when the threshold exceeds 90%. Given that the accuracy of flood and nonflood extraction is considered sufficient when the commission is below 1%, we designate 90% as the ideal threshold and classify the SAR images into three groups.

2) Classification of Uncertainty Region: The traditional categorization methods typically rely on absolute binary division, whereas fuzzy logic offers a more flexible approach. It quantifies the degree to which items belong to different categories based on membership values [37], [55], thus improving both the accuracy and stability of flood mapping. In the procedure described, the SAR image is initially divided into three groups based on probability thresholds, with uncertain regions requiring further classification. Accurate classification of these uncertain areas cannot be achieved solely based on SAR gray values due to feature mixing and noise effects. In SAR images, texture features are independent of the magnitude of the gray values and describe the repetition or variation between image pixels, being derived from the gray covariance matrix. Hence, texture features emerge as another essential factor for flood extraction [10], [56]. In this study, the fuzzy set for classifying uncertain regions incorporates both SAR gray values and texture features. Specifically, the angular second-order moment (ASM) is the only texture feature considered in this article. ASM measures both the uniformity of pixel grayscale distribution and the fineness of the image texture, with higher values indicating a more uniform gray distribution and finer texture.

The fuzzy membership functions utilized include the S and Z functions. Flood areas in SAR images generally correspond to low gray values, which aligns with the Z function, as the membership increases with lower input values. The Z function is parameterized as follows:

$$a_Z = x_{p=90\%}, \quad c_Z = x_{p=10\%} \tag{4}$$

where $x_{p=90\%}$ and $x_{p=10\%}$ represent gray values with probability thresholds of 90% and 10%, respectively.

Conversely, high ASM values are typically indicative of flooded areas. The S-function represents fuzzy membership for ASM, as it shows a higher degree of membership when the input value is high, consistent with the flood exhibiting high ASM values. The S function is parameterized as

$$a_S = \text{mean}_{\text{ASM}} - \text{std}_{\text{ASM}}, \quad c_S = \text{mean}_{\text{ASM}} + \text{std}_{\text{ASM}} \tag{5}$$

where mean_{ASM} and std_{ASM} denote the mean and standard deviation of the initial classification of flood ASM. The uncertainty region is classified by averaging the two fuzzy memberships, with regions considered flooded if the average membership is $\geq$0.5, and nonflooded if it is $\leq$0.5. Multiscale segmentation of images often results in ambiguity and instability due to factors such as parameter settings, the

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TABLE II
COMMISSIONS (%) FOR WATER AND NONWATER AT VARIOUS PROBABILITY THRESHOLDS

SAR image	probability / area	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%
Aug.18, 2016	flood	5.26	4.52	3.83	3.20	2.66	2.23	1.87	1.23	**0.89**	0.56
	non-flood	1.23	1.21	1.15	1.09	1.04	1.00	0.96	0.93	**0.90**	0.87
Jul.20,2017	flood	5.58	5.55	4.89	4.11	3.38	2.33	1.93	1.34	**0.94**	0.45
	non-flood	1.29	1.14	1.06	1.02	0.97	0.93	0.89	0.84	**0.81**	0.78
Jul.27,2018	flood	4.56	3.87	3.39	2.85	2.35	2.04	1.67	1.15	**0.81**	0.45
	non-flood	1.43	1.33	1.31	1.21	1.14	1.07	1.02	0.94	**0.92**	0.88
Aug.3,2019	flood	7.96	6.84	5.87	4.35	3.76	2.71	1.72	1.22	**0.61**	0.20
	non-flood	1.53	1.41	1.28	1.20	1.04	1.00	0.96	0.88	**0.85**	0.79
Aug.21, 2020	flood	3.91	3.80	3.28	2.77	2.12	1.77	1.38	0.90	**0.51**	0.20
	non-flood	1.21	1.14	1.09	1.05	1.00	1.00	0.98	0.94	**0.93**	0.89
Jul.23,2021	flood	5.79	4.89	4.04	3.33	2.61	2.08	1.66	1.01	**0.68**	0.35
	non-flood	1.21	1.20	1.13	1.06	1.00	0.96	0.94	0.91	**0.87**	0.82

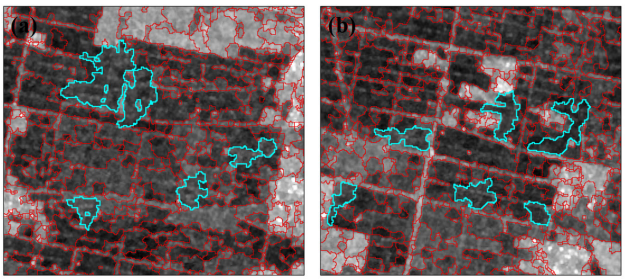


Fig. 10. Example of multiscale segmentation results, where the highlighted regions represent areas with inaccurate segmentation boundaries.

inherent complexity of the image, and noise interference. Two schematics of the multiscale segmentation results are provided in Fig. 10. We can observe that although the segmentation results divide the image into different regions more clearly, some boundaries remain inaccurate, which negatively impacts subsequent image analysis and processing tasks. It is a typical challenge with object-level methods. Therefore, the study combines object- and pixel-level methods. Before classifying the uncertain region, the standard deviation of image elements within each object is calculated. If the standard deviation of an object exceeds the total standard deviation of the pre-classified flood regions, the object is converted into pixel level for further classification; otherwise, classification continues at the object level.

IV. RESULTS AND DISCUSSION

A. Accuracy Assessment

This study compares the proposed algorithm with several widely used flood monitoring methods to evaluate its performance. Additionally, the algorithm employs a two-step classification process, integrating fuzzy logic and object-pixel combined strategies in the second step to classify uncertain regions and enhance the accuracy of flood mapping. The performance of the two-step classification algorithm is further validated through a comparative analysis. Four standard evaluation metrics—recall, precision, OA, and $F1$ score—are employed to evaluate the accuracy of the flood mapping outcomes. Recall measures the likelihood that a pixel identified as flood actually corresponds to flood in reality, highlighting the model's effectiveness in identifying flood regions. Precision, on the other hand, is the proportion of pixels predicted to be floods that are correctly identified as such, assessing the accuracy of the model in flood detection. The $F1$ score, calculated as the harmonic mean of recall and precision, provides a balanced measure of the model's performance in both areas. OA serves as an overall measure, evaluating the model's general ability to correctly classify pixels as either flood or nonflood

$$\begin{aligned} \text{Recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{Precision} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ F1 &= \frac{2\text{recall}\times\text{Precision}}{\text{recall}+\text{Precision}} \\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{FN}+\text{TN}} \end{aligned} \tag{6}$$

where true positive (TP) represents the number of pixels that are genuinely flood and correctly identified as such, false negative (FN) refers to the number of pixels that are truly flood but incorrectly classified as nonflood, true negative (TN) refers to the number of pixels correctly identified as nonflooded, and false positive (FP) indicates the number of pixels that are not floods but are mistakenly classified as floods.

1) Comparison With Existing Algorithms: Our algorithm integrates fuzzy logic, the Gaussian mixture model, and OBIA methods. To validate its performance, we compared it with several well-established and highly accurate algorithms that encompass one or more of these techniques. The comparison algorithms include the fully automatic flood process (FAFP) [36], [37], normalized difference flood index (NDFI) [11], rapid flood processor (RFP) [55], and the improved Gaussian mixture model (IGMM) [29]. SAR images of typhoon-induced floods in the study area from 2016 to 2021 were downloaded, with representative flood images selected for each year. The proposed method and comparative methods were then applied to extract flood areas, and the results were analyzed and compared. Figs. 11 and 12 show the flood areas and associated error distributions, estimated by different algorithms based on

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TABLE III
QUANTITATIVE ACCURACY ASSESSMENT OF FLOOD MAPPING USING DIFFERENT ALGORITHMS BASED ON SAR DATA FROM AUGUST 18, 2016, AND JULY 27, 2018

SAR image	Methods	Recall	Precision	F1	OA
Aug.18,2016	our method	89.33%	90.09%	**89.71%**	**98.33%**
	FAFP	53.92%	**95.11%**	68.83%	96.01%
	NDFI	66.74%	83.02%	74.00%	96.17%
	RFP	91.69%	72.80%	81.59%	96.52%
	IGMM	**97.66%**	37.24%	53.96%	86.38%
Jul.27,2018	our method	96.25%	81.65%	**88.35%**	**97.94%**
	FAFP	52.51%	**96.19%**	67.93%	95.98%
	NDFI	83.58%	66.32%	73.96%	95.23%
	RFP	7394%	82.84%	78.13%	96.64%
	IGMM	**96.62%**	51.86%	67.49%	92.46%

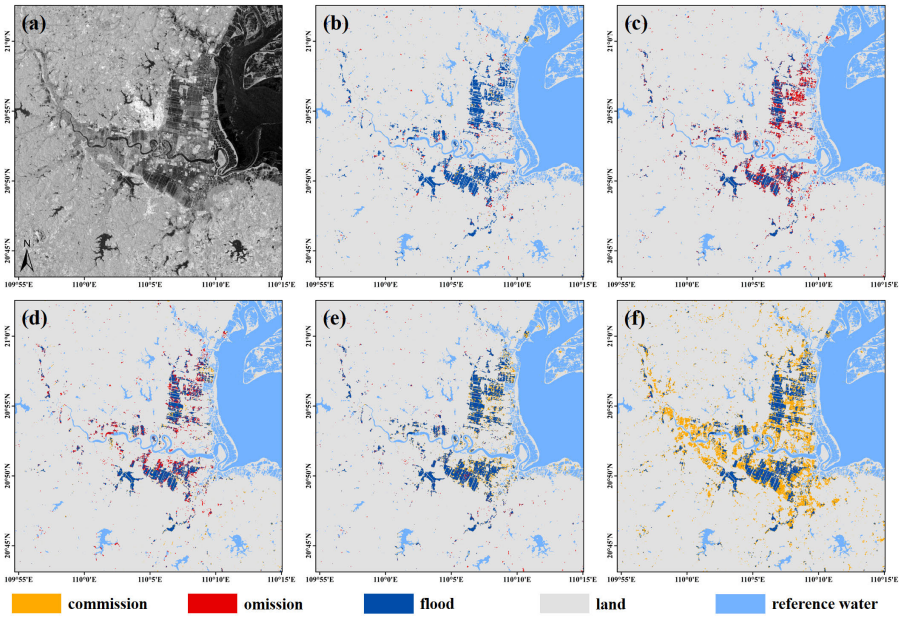


Fig. 11. Estimated flood areas and error distributions based on SAR data from August 18, 2016. (a) SAR image. (b) Proposed method. (c) FAFP. (d) NDFI. (e) RFP. (f) IGMM.

SAR images from August 18, 2016, and July 27, 2018, respectively, with the corresponding assessment indicators provided in Table III. Table IV provides the average performance of each algorithm from 2016 to 2021. The comparative analysis reveals that the FAFP method achieves the highest precision but also suffers from a low recall. In contrast, the IGMM method shows the highest recall rate but the lowest precision rate. The primary reason for this difference lies in the reliance of both methods on single-polarization data, which often introduces greater errors in flood mapping. This observation aligns with the findings discussed in Section III-B, further highlighting the limitations of using single-polarization data. While achieving high precision or recall is valuable, neither metric alone is sufficient to evaluate the overall effectiveness of the algorithm. High precision ensures that most detected areas are correct, but if recall is low, a significant portion of relevant areas remains undetected, which could be critical in applications like disaster response. Conversely, high recall ensures comprehensive detection but may come at the cost of numerous false positives, reducing the reliability of the results. Therefore, striking a balance between precision and recall is essential. Metrics like the $F1$ score and OA, which harmonize these two aspects, provide a more comprehensive evaluation of an algorithm's robustness and practical utility. Notably, the method proposed in this study achieves neither the highest precision nor the highest recall but outperforms other methods in terms of $F1$ score and OA, highlighting its superior balance and overall effectiveness.

TABLE IV
AVERAGE QUANTITATIVE ACCURACY ASSESSMENT OF FLOOD MAPPING FOR DIFFERENT ALGORITHMS BETWEEN 2016 AND 2021

Methods	Recall	Precision	F1	OA
our method	91.31%	81.10%	**85.53%**	**97.79%**
FAFP	48.86%	**91.45%**	63.17%	96.05%
NDFI	70.74%	67.04%	67.30%	95.59%
RFP	73.72%	68.82%	68.92%	95.05%
IGMM	**96.18%**	42.73%	58.57%	90.43%

2) Evaluation of Two-Step Classification: A two-step classification method was employed in the flood mapping process to enhance categorization accuracy. In the second step,

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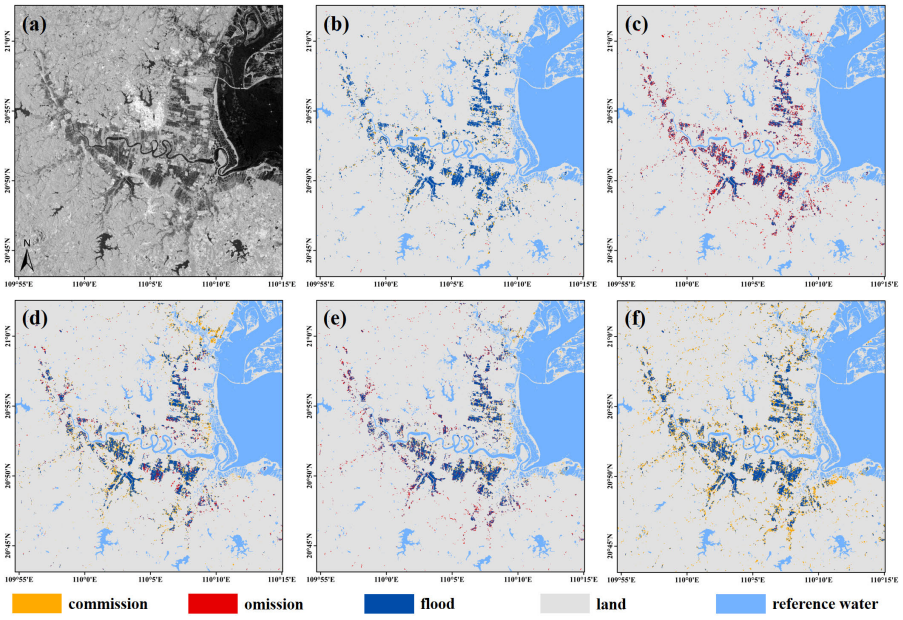


Fig. 12. Estimated flood areas and error distributions based on SAR data from July 27, 2018. (a) SAR image. (b) Proposed method. (c) FAFP. (d) NDFI. (e) RFP. (f) IGMM.

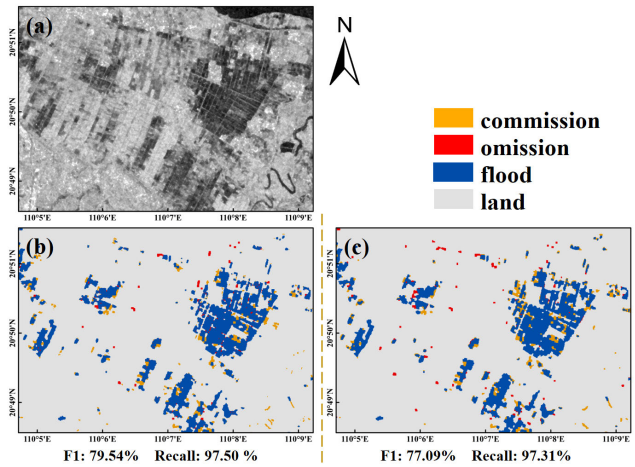


Fig. 13. (a) SAR image. (b) and (c) Estimated flood areas and error distributions based on the combination of grayscale values and the ASM fuzzy system, as well as those based solely on grayscale values.

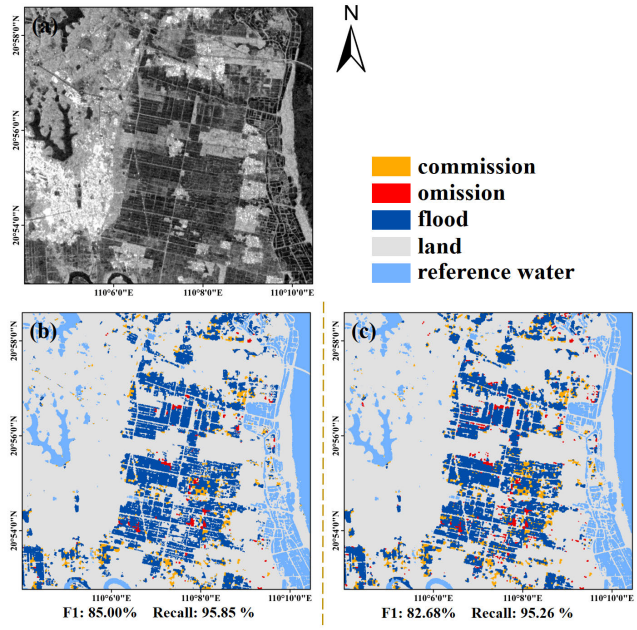


Fig. 14. (a) SAR image. (b) Estimated flood areas and error distributions using the object-pixel combined method. (c) Estimated flood areas and error distributions using the object-level method.

classification was refined using fuzzy sets incorporating texture features and gray values. To validate its effectiveness and superiority, this method was evaluated against the grayscale-based method described by [29]. Both methods were tested using VV + VH SAR images under optimal conditions to address potential issues arising from different polarization data. As outlined in [29], their method utilized GMM to convert P-values into binary flood maps for crops, using a 50% threshold. The error distributions of the two methods are presented in Fig. 13. Compared to the grayscale-based method, the fuzzy set-based approach, which integrates gray values and ASM, demonstrated lower omission and commission errors, resulting in more accurate flood mapping.

In the flood mapping process, an object-pixel combined approach was applied to mitigate inaccuracies caused by multiscale segmentation. The effectiveness of this method was assessed by comparing it with the object-level method. Error distribution maps for both methods are presented in Fig. 14. Compared with the object-level method, the object-pixel combined approach significantly reduces both commission and omission errors with high $F1$ and recall. These results demonstrate that the combined method enhances the accuracy of flood mapping and effectively mitigates errors caused by segmentation.

B. Flood Analysis

To predict the risk of flooding in the study area, we conducted an analysis of the relationship between rainfall and

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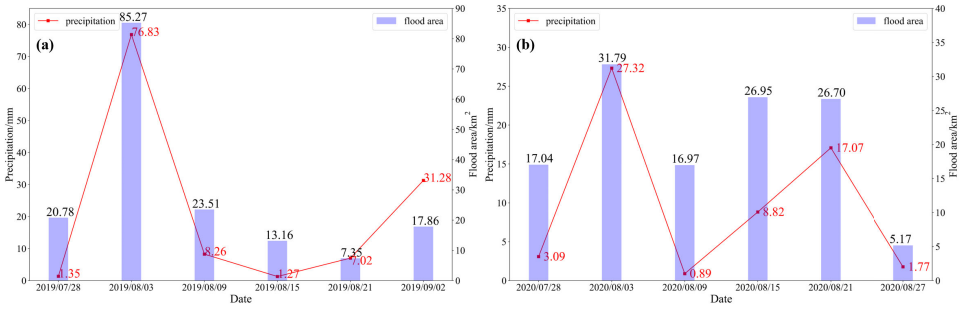


Fig. 15. Relationship between rainfall and flooded areas during July and August in (a) 2019 and (b) 2020.

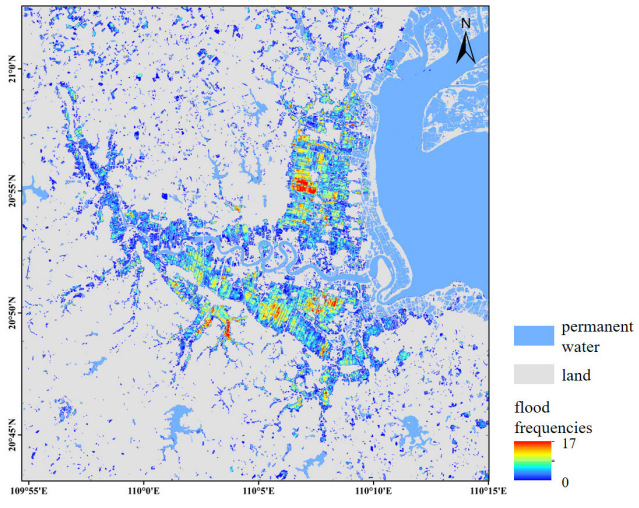


Fig. 16. Flood frequencies of the study area.

flooded areas. The rainfall data were sourced from the global station-based observational dataset released by the National Centers for Environmental Information (NCEI) under the National Oceanic and Atmospheric Administration (NOAA), covering the period 1929–2024. Using daily rainfall records from all stations in China, we generated nationwide daily rainfall raster data at a 1-km spatial resolution through inverse distance weighting interpolation. Given that the region experienced multiple flood events during the typhoon seasons of late July to August in 2019 and 2020, which were recorded by Sentinel-1 satellites, we focused on these two periods to analyze the correlation between rainfall and flood. Flooding is influenced not only by the rainfall on the day of the event but also by the cumulative rainfall from the previous two days. Significant rainfall in the preceding days increases soil saturation, meaning that even moderate rainfall on the current day will lead to flooding if the soil cannot absorb additional water. Therefore, we compared the average rainfall on the day of SAR image acquisition and the two preceding days with the corresponding flooded areas. Fig. 15(a) and (b) shows the comparison between average rainfall and flooded areas for 2019 and 2020, respectively. The data analysis clearly indicates that, in most cases, higher rainfall corresponds to larger flooded areas. However, there are also some special cases, such as a small amount of rainfall resulting in a large flooded area or heavy rainfall causing only a small flooded area. The former may occur when two flood events are closely spaced in time, with the previous floodwater not fully receded, while the latter may be due to dry soil conditions before the flood, which absorb a significant portion of the rainfall. Overall, rainfall provides a general indication of floods, offering valuable reference information for flood monitoring and early warning systems.

To further investigate the frequency of flood disasters in the region, this study conducted a statistical analysis of 19 Sentinel-1 SAR images affected by flooding between 2016 and 2021. First, the flood monitoring algorithm proposed in this study was used to identify flooded regions in the SAR images. Subsequently, the flood frequency map was generated. Fig. 16 illustrates that the largest flooded area reached 85.27 km^2, representing 15% of the total cultivated land in the region. Additionally, in the 19 Sentinel-1 SAR images, 27.80 km^2 of land has been flooded more than ten times, representing high-risk zones for flood disasters that warrant special attention.

V. Conclusion

The complexity of surface features and the richness of scattering information pose significant challenges in accurately distinguishing various land cover types using single-polarization SAR data. Through experimental analysis using Sentinel-1 SAR, we confirmed the pivotal role of integrating dual-polarization information in improving classification accuracy and reliability under diverse land cover and vegetation conditions. Building on these findings, this study introduces a high-precision flood mapping method utilizing Sentinel-1 dual-polarization SAR imagery, effectively addressing the limitations of single-polarization data and overcoming the challenges associated with complex surface features in vegetated regions. By incorporating Sentinel-1 polarization data into the multiscale segmentation process, the method significantly enhances the quality of segmentation results. By analyzing the GMM overlap areas under different polarization modes, we demonstrated that SAR images derived from dual-polarization additive combinations substantially reduce the impact of feature mixing. The flood mapping process is implemented using a two-step classification strategy: the first step performs an initial classification of the entire area, while the second step refines the results by focusing on uncertain regions. This step integrates two fuzzy sets based on SAR

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intensity values and texture features to achieve a more refined classification. Additionally, dynamic adjustments at both pixel and object levels are implemented to mitigate errors stemming from insufficient segmentation precision, enhancing the reliability of classification results. This study systematically analyzed the frequency and spatial extent of flooding in the study area from 2016 to 2021, revealing frequent flood events triggered by typhoons and heavy rainfall. Experimental comparisons indicate that the proposed method surpasses existing algorithms, achieving a more favorable balance between omission and commission errors, as demonstrated by the $F1$ score of 85.53% and the OA of 97.79%. Overall, the high-quality Sentinel-1 data form a reliable basis for implementing precise algorithms, greatly improving the accuracy and timeliness of flood disaster monitoring, and providing essential technical support for disaster management.

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